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|---|-----|-------|---|----------------------|-----|
| 3 | (a) | (i) | <u>amplitude</u> remains constant | B1 | [1] |
| | | (ii) | <u>amplitude</u> decreases gradually
light damping | M1
A1 | [2] |
| | | (iii) | period = 0.80 s
frequency = 1.25 Hz (<i>period not 0.8 s, then 0/2</i>) | C1
A1 | [2] |
| | (b) | (i) | (induced) e.m.f. is proportional to
rate of change/cutting of (magnetic) flux (linkage) | M1
A1 | [2] |
| | | (ii) | a current is induced in the coil
as magnet moves in coil
current in resistor gives rise to a heating effect
thermal energy is derived from energy of oscillation of the magnet | M1
A1
M1
A1 | [4] |
| 4 | (a) | (i) | zero field (strength) inside spheres | B1 | [1] |